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Mohammad Miadul Islam Jasim — ETB 2021

Chapter – 1

Introduction to Web Design and HTML

1.1 Core Concepts in Web Design

Internet

Internet is the connection of millions of computers worldwide that allows them to communicate and share information.

World Wide Web

World Wide Web is a system of sharing information over the Internet by using hypertext documents, which are linked together and can be accessed through a web browser.

Note: The Internet is the hardware, wires, and protocols—the global infrastructure. The World Wide Web is a service (like email or gaming) that runs on top of that infrastructure.

Hypertext document file

A hypertext document is a type of document that contains links to other documents, allowing users to navigate between them easily.

- Like images this is a file type with extension .html or .htm.
- It is made by using HTML (HyperText Markup Language).

Webpage

A webpage is a single hypertext document or page that is displayed in a web browser. It can contain text, images, links of other webpages, and other multimedia content.

Website

A website is a collection of related webpages stored under a common domain name and connected through hyperlinks.

Web design

Web design is the process of planning, creating, arranging and presenting information on the World Wide Web. It includes the design of webpages, navigation system, content layout, links, images, and user interaction.

1.1.1 Basic Relationship

- A webpage is one single page.
- A website contains multiple related webpages.
- The first or main page of a website is called the **homepage**.
- Webpages are connected using hyperlinks.
- A web browser is used to view webpages.

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Core Idea

HTML → Hypertext document files → Webpage $\xrightarrow{\text{Web Design}}$ Website → World Wide Web → Internet

1.1.2 Advantages of Website

- Information can be published and accessed easily.
- A website can be accessed from anywhere in the world.
- It is available twenty-four hours a day.
- It reduces the need for physical communication.
- It supports text, images, audio, video and links.
- Information can be updated more easily than printed documents.
- It is useful for education, business, news, communication and entertainment.

1.2 IP Address

IP Address

IP Address stands for Internet Protocol Address. It is a unique numerical address used to identify a device connected to a network on the internet.

Note: An IP address works like the digital address of a device. When sender sends data to a receiver, the receiver's IP address is used to ensure that the data reaches the correct destination.

1.2.1 Purpose of IP Address

- It identifies a device in a network.
- It helps send data to the correct destination.
- It allows communication between client and server.
- It is necessary for internet-based communication.

1.2.2 Types of IP Address

→ There are two main types of IP addresses:

- **IPv4**: 32-bit address, with four 8-bit octets, 1 octet = 8 bits, usually written in four decimal parts.
  Example: `11000000.10101000.00000001.00000001` $\xrightarrow{\text{in decimal}}$ `192.168.1.1`
- **IPv6**: 128-bit address, with eight 16-bit words, usually written in hexadecimal format.
  Example: `2001:0db8:85a3:0000:0000:8a2e:0370:7334`

1.2.2.1 Network ID and Host ID

- The internet is a network of many networks.
- Each network has many hosts (computers, mobiles, servers etc.) in it.
- So each network and each device needs an unique identifier for communication.
- This is done by dividing the IP address into two parts: Network ID and Host ID.

→ **Network ID**: identifies the network the device is connected to.
Example: in `192.168.1.1`, the network ID is `192.168.1`.

→ **Host ID**: identifies the specific device inside that network.
Example: in `192.168.1.1`, the host ID is `1`.

Example 1.1

Q: Why is IP address required?

A: IP address is required because every device in a network must be uniquely identified. It helps data packets travel from the sender to the correct receiver.

Conclusion: Without IP addresses, devices on the internet could not communicate properly.

1.3 Domain Name

Domain Name

A domain name is a human-readable name used to identify a website on the internet.

Note: IP addresses are difficult for humans to remember. For this reason, domain names are used instead of numeric IP addresses.

1.3.1 Need for Domain Name

- It makes website addresses easier to remember.
- It hides the complexity of numeric IP addresses.
- It gives a website a readable identity.
- It helps users access websites easily.

DNS

DNS stands for Domain Name System. It converts a domain name into its corresponding IP address.

1.3.2 Example of Domain Name Structure

Domain Name Structure

$$www.example.com$$

$$www = \text{Host Name}$$

$$example = \text{Second Level Domain}$$

$$.com = \text{Top Level Domain}$$

1.3.3 Common Top Level Domains

Domain	Used For
.com	Commercial organizations
.net	Network-related organizations or services
.edu	Educational institutions
.gov	Government organizations
.org	Non-profit organizations
.mil	Military organizations

Table 1.1: Common Top Level Domains

Domain Name vs IP Address

- IP Address is used by computers.
- Domain Name is used by humans.
- DNS connects the two by converting domain names into IP addresses.

1.4 URL (Uniform Resource Locator)

URL

A URL (Uniform Resource Locator) is the complete address used to locate a resource on the internet.

A URL not only identifies the location of a webpage but also specifies the method used to access it.

1.4.1 Structure of URL

General Form

$$\text{URL} = \text{Protocol} + \text{Domain Name} + \text{Path}$$

Example: $https://www.example.com/folder/page.html$

1.4.2 Components of URL

- **Protocol**: Defines the communication rule (e.g., HTTP, HTTPS).
- **Domain Name**: Identifies the server.
- **Path**: Specifies the exact location of the resource in the server.

Summary

A URL uniquely identifies both the location and access method of a web resource.

1.5 Website Structure

Website Structure

Website structure refers to the way webpages are organized and interconnected within a website.

A proper structure ensures easy navigation and better user experience.

1.5.1 Types of Website Structure

1.5.1.1 Hierarchical Structure

- Pages are arranged like a tree.
- Starts from a homepage and branches into sections.
- Each section may contain sub-sections.
- Easy to manage and most commonly used.

1.5.1.2 Linear Structure

- Pages are connected in a sequence.
- Users move step-by-step (next or previous).
- Suitable for tutorials or guided content.

1.5.1.3 Network Structure

- Each page is connected with many other pages.
- No fixed navigation path.
- Provides flexibility but may confuse users.

1.5.1.4 Hybrid Structure

- Combination of multiple structures.
- Uses hierarchical with linear or network links.
- Most modern websites follow this model.

Summary

A good website structure improves navigation, usability, and overall performance.

1.6 Types of Website

1.6.1 Static Website

Static Website

A static website displays fixed content that does not change unless manually updated.

- Built using HTML and CSS.

- Content remains the same for all users.
- Faster loading speed.
- Simple to develop.

1.6.2 Dynamic Website

Dynamic Website

A dynamic website generates content dynamically based on user interaction or database.

- Uses server-side scripting (e.g., PHP, ASP, Java).
- Content can change for different users.
- Supports user interaction.
- More complex than static websites.

1.6.3 Comparison

Feature	Static Website	Dynamic Website
Content	Fixed	Changeable
Update	Manual	Automatic / dynamic
Speed	Faster	Slower
Technology	HTML, CSS	Server-side languages
Interaction	Not supported	Supported
Complexity	Simple	Complex

Table 1.2: Comparison between Static and Dynamic Website

Summary

Static websites are simple and fast, whereas dynamic websites are flexible and interactive.

1.7 Web Browser and Search Engine

1.7.1 Web Browser

Web Browser

A web browser is a software application used to access, retrieve, and display webpages.

- It interprets HTML and renders content visually.
- It allows users to navigate between webpages.
- Examples include Google Chrome, Mozilla Firefox, Microsoft Edge, Opera.

1.7.2 Search Engine

Search Engine

A search engine is a system that helps users find information on the internet by searching indexed webpages.

- It collects and organizes web content.
- It returns relevant results based on user queries.
- Examples include Google, Bing, Yahoo.

1.7.3 Difference Between Browser and Search Engine

- A browser is used to open and view websites.
- A search engine is used to find websites.

Summary

Browser is a tool for accessing the web, while a search engine is a tool for searching information.

1.8 HTML (HyperText Markup Language)

HTML

HTML (HyperText Markup Language) is the standard markup language used to create and structure webpages.

HTML does not perform computation; it describes the structure and content of a webpage so that a browser can render it.

1.8.1 Key Characteristics

- Uses tags to mark content.
- Works with web browsers to display pages.
- Supports text, images, links, lists, tables and forms.
- Platform independent (runs in any modern browser).

1.8.2 Advantages of HTML

- Simple and easy to learn.
- Supported by all browsers.
- Integrates with CSS and JavaScript.
- Lightweight and fast to load.

1.8.3 Limitations of HTML

- Cannot create dynamic behavior by itself.

- Limited styling without CSS.
- No database interaction.

Summary

HTML defines structure; CSS handles styling; JavaScript adds interactivity.

1.9 Basic Structure of an HTML Document

HTML Document Structure

An HTML document is organized using a standard structure consisting of `<!DOCTYPE>`, `<html>`, `<head>` and `<body>`.

1.9.1 Skeleton Code

This example shows the minimum structure of a valid HTML page. The browser uses `<!DOCTYPE html>` to understand that the document follows HTML5 rules. The `head` stores page information, while the `body` stores visible content.

```
<!DOCTYPE html>
<html>
<head>
    <title>Page Title</title>
</head>
<body>
    Content goes here
</body>
</html>
```

Listing 1.1: Basic HTML Document Structure

What to notice

- The `html` element contains the whole page.
- The `title` appears in the browser tab.
- Visible page content is written inside `body`.

1.9.2 Explanation of Parts

- `<!DOCTYPE html>` declares the document type (HTML5).
- `<html>` is the root element.
- `<head>` contains metadata (title, links, scripts).
- `<title>` sets the browser tab title.
- `<body>` contains visible content.

Summary

The `<head>` holds information about the page, while the `<body>` holds what users see.

1.10 HTML Tags

Tag

A tag is a keyword enclosed in angle brackets used to define elements in HTML.

1.10.1 Types of Tags

- **Container (Paired) Tags**: Have opening and closing parts. Example: `<p> ... </p>`
- **Empty (Unpaired) Tags**: Do not have a closing tag. Example: `<br>`, `<hr>`

1.10.2 Examples

This example compares paired tags and empty tags. A paragraph uses both opening and closing tags, but `br` and `hr` are empty elements.

```
<p>This is a paragraph</p>
<br>
<hr>
```

Listing 1.2: Paired and Empty HTML Tags

What to notice

- `p` is a container tag because it wraps content.
- `br` creates a line break without a closing tag.
- `hr` creates a horizontal line.

Summary

Tags define how content is structured and displayed in a webpage.

1.11 HTML Elements

Element

An HTML element consists of a start tag, content, and an end tag.

1.11.1 Structure of an Element

Formula

$$\text{Element} = \text{Start Tag} + \text{Content} + \text{End Tag}$$

1.11.2 Example

This short example shows one complete HTML element. The start tag, content, and end tag together form the element.

```
<p>Hello World</p>
```

Listing 1.3: A Complete Paragraph Element

What to notice

- The content is written between the two tags.
- The closing tag uses a slash before the tag name.

- `<p>`: Start tag
- `Hello World`: Content
- `</p>`: End tag

Summary

Tags are the markers, while elements are the complete units formed by tags and content.

1.12 HTML Attributes

Attribute

An attribute provides additional information about an HTML element and is written inside the start tag.

1.12.1 General Form

Formula

$$\text{<tag attribute="value">}$$

1.12.2 Examples

This example shows how attributes add extra information to HTML elements. The `href` attribute gives a link destination, while `src` and `alt` describe an image source and fallback text.

```
<a href="https://www.example.com">Visit</a>
<img src="image.jpg" alt="Image">
```

Listing 1.4: Common HTML Attributes

What to notice

- Attributes are written inside the opening tag.
- Attribute values are usually written inside quotation marks.
- The same element can have more than one attribute.

1.12.3 Common Attributes

- `href`: defines link destination
- `src`: defines image source

- `alt`: alternative text for images
- `title`: tooltip text

Summary

Attributes modify the behavior or appearance of elements.

1.13 Text Formatting Tags

1.13.1 Basic Formatting

Tag	Purpose	Example
<b>	Bold text	`<b>Text</b>`
<i>	Italic text	`<i>Text</i>`
<u>	Underline text	`<u>Text</u>`
<sup>	Superscript	`a<sup>2</sup>`
<sub>	Subscript	`H<sub>2</sub>O`
 	Line break	`Line Next`
<hr>	Horizontal line	`<hr>`

Table 1.3: Common HTML Formatting Tags

Summary

Formatting tags are used to control the appearance of text.

1.14 HTML Lists

HTML provides different types of lists to organize content.

1.14.1 Ordered List (OL)

Ordered List

An ordered list displays items in a numbered sequence.

This example creates a numbered list. Each list item must be written inside an `li` element.

```
<ol>
    <li>Item 1</li>
    <li>Item 2</li>
</ol>
```

Listing 1.5: Ordered List

What to notice

- `ol` means ordered list.

- Browsers number ordered list items automatically.

1.14.2 Unordered List (UL)

Unordered List

An unordered list displays items using bullet points.

This example creates a bullet list. It is useful when the order of items is not important.

```
<ul>
    <li>Item A</li>
    <li>Item B</li>
</ul>
```

Listing 1.6: Unordered List

What to notice

- `ul` means unordered list.
- Each bullet is created by an `li` tag.

1.14.3 Description List (DL)

Description List

A description list is used to define terms and their descriptions.

This example creates a description list. It pairs a term with a short explanation.

```
<dl>
    <dt>HTML</dt>
    <dd>Markup language</dd>
</dl>
```

Listing 1.7: Description List

What to notice

- `dt` defines the term.
- `dd` gives the description of that term.

Summary

HTML lists help organize content in a structured and readable way.

1.15 Images in HTML

Image Tag

The `<img>` tag is used to display images on a webpage.

1.15.1 Basic Syntax

This example inserts an image into a webpage. The `src` attribute gives the image file path, and `alt` provides text if the image cannot load.

```
<img src="image.jpg" alt="Description">
```

Listing 1.8: Basic Image Tag

What to notice

- `img` is an empty element.
- `alt` improves accessibility and helps when the image path is wrong.

1.15.2 Common Attributes

- `src`: specifies image path
- `alt`: alternative text if image fails to load
- `width`: image width
- `height`: image height

Summary

Images enhance visual presentation and improve user experience.

1.16 Hyperlinks in HTML

Hyperlink

A hyperlink is a connection that allows navigation from one webpage to another.

1.16.1 Anchor Tag

This example creates a clickable link. The `href` attribute stores the destination address, and the text between the tags becomes clickable.

```
<a href="https://www.example.com">Visit Website</a>
```

Listing 1.9: Basic Hyperlink

What to notice

- The `a` tag creates an anchor.
- The link destination is controlled by `href`.

1.16.2 Important Attributes

- `href`: destination URL
- `target`: where to open link

1.16.3 Target Values

- `_self`: opens in same tab
- `_blank`: opens in new tab

Summary

Hyperlinks are the foundation of web navigation, connecting pages together.

1.17 HTML Tables

Table

An HTML table is used to display data in rows and columns.

1.17.1 Basic Structure

This example builds a small table using rows, header cells, and data cells. The first row contains headings, and the second row contains normal table data.

```html
<table border="1">
    <tr>
        <th>Name</th>
        <th>Age</th>
    </tr>
    <tr>
        <td>John</td>
        <td>20</td>
    </tr>
</table>
```

Listing 1.10: Basic HTML Table

What to notice

- `tr` creates a table row.
- `th` creates a heading cell.
- `td` creates a normal data cell.

1.17.2 Table Elements

- `<table>`: defines table
- `<tr>`: table row
- `<th>`: table header cell
- `<td>`: table data cell

1.17.3 Common Attributes

- `border`: defines border size
- `cellpadding`: space inside cells
- `cellspacing`: space between cells

Summary

Tables are used to present structured data clearly in rows and columns.

1.18 Final Summary of the Chapter

Chapter Overview

- Web design deals with creating and structuring websites.
- Websites are collections of interconnected webpages.
- IP address uniquely identifies devices on a network.
- Domain name provides a human-readable address.
- URL specifies the location of resources.
- Website structure affects navigation and usability.
- HTML is used to create webpage structure.
- Tags, elements, and attributes form HTML content.
- Lists, images, links, and tables are essential HTML components.

Mohammad Mizadul Islam Jasim - 2108033